

Prompt Injection and Tool-Use Security in LLM Agents: A Survey

Prompt injection and tool-use security define a central risk frontier for LLM agents. When models consume untrusted text, retrieve documents, browse websites, or call external tools, instructions can be smuggled through data channels and converted into actions, leakage, or policy violations. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around direct and indirect prompt injection, jailbreaks, tool and function-call permissions, data exfiltration, RAG poisoning, guardrails, monitoring, red teaming, and alignment. The survey argues that security for LLM agents is a systems problem: models, prompts, retrieval stores, tools, policies, sandboxes, and users form one attack surface. We close with prospects for capability-scoped tools, provenance-aware context, robust instruction hierarchies, continuous monitoring, and evaluation protocols that measure real trajectories rather than only single-turn refusals.

Keywords:

large language models; prompt injection; tool-use security; LLM agents; jailbreaks; guardrails; privacy

1 Introduction

LLM agents are useful because they connect language models to tools, documents, websites, memories, databases, and external actions. The same connectivity creates a new security problem: text that appears to be data can become an instruction, and model-generated text can become an action. Prompt injection exploits this ambiguity by placing malicious instructions in a channel the model reads, while tool-use attacks exploit the model’s ability to call APIs, browse web pages, write files, or transmit information [5, 15, 44, 65, 167, 168, 220, 256]. Security therefore cannot be treated as a final output filter; it must be designed into the agent loop.

The threat becomes sharper in indirect prompt injection. A user may ask an agent to summarize a web page, email, PDF, or retrieved document, and the untrusted content may contain instructions to ignore prior constraints, reveal secrets, or invoke tools. If the agent fails to separate data from instructions, the attacker can influence actions without directly communicating with the model operator [5, 15, 44, 65, 167, 168, 220, 256]. Tool-rich agents

add additional consequences: a compromised trajectory may send messages, modify records, exfiltrate context, or authorize unsafe operations.

This survey treats prompt injection and tool-use security as a full-stack problem. The model, prompt hierarchy, retrieval system, tool schema, permission boundary, memory store, monitor, and human approval interface all shape the attack surface. The main question is not whether an LLM can refuse a malicious sentence in isolation, but whether an agent can preserve user intent and policy constraints while operating over untrusted inputs and external capabilities.

2 Background

2.1 Instruction/Data Confusion

Traditional software separates code and data through parsers, type systems, and privilege boundaries. LLM agents blur this separation because the same text channel may contain user goals, system policies, retrieved evidence, tool outputs, and environment observations. Prompt injection is best un-

derstood as instruction/data confusion: the model treats adversarial data as higher-priority instructions. The problem is amplified by long contexts, retrieval, and web interaction, where the origin and authority of each text span may be unclear [5, 15, 44, 65, 167, 168, 220, 256].

2.2 Tool Use as a Capability Boundary

A tool call is a capability transfer. Function calling exposes a structured action space, but schemas do not automatically encode permission, intent, reversibility, or data-flow constraints. A safe tool-using agent needs to know what a tool can do, what data it can access, which calls are irreversible, and whether the current user authorized the action. Tool-use security therefore connects LLM prompting with classical concepts such as least privilege, sandboxing, taint tracking, capability tokens, and audit logs [5, 15, 44, 65, 167, 220, 256, 265].

2.3 From Jailbreaks to Agentic Attacks

Jailbreaks typically try to elicit prohibited text. Agentic attacks may instead aim to produce a harmful action, leak a secret, poison memory, corrupt retrieval, or manipulate a workflow. This difference matters for evaluation. A model can refuse direct harmful requests yet still follow malicious instructions embedded in a document, or it can generate benign text while silently calling an unsafe tool. Agent safety must therefore be measured over trajectories [5, 15, 65, 167, 168, 220, 256, 265].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the topic with prompt injection, indirect prompt injection, jailbreaks, prompt attacks, LLM security, tool-use security, function-calling security, data exfiltration, prompt leakage, untrusted content, web-agent security, RAG poisoning, guardrails, red teaming, adversarial prompts, policy compliance, misuse, and sandbox terms. Records were retained when title or abstract evidence connected LLMs or language models with attacks, defenses, privacy, agent security, tool permissions, alignment, evaluation, or robustness. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the field along six axes. The first is *attack channel*: direct user prompt, retrieved document, web page, tool output, memory, file, or multimodal content. The second is *target*: model behavior, tool call, secret, memory, retrieval index, policy, or human decision. The third is *capability*: read-only answering, browsing, writing, executing code, sending messages, or modifying external state. The fourth is *defense layer*: prompt hierarchy, context provenance, tool permissions, sandboxing, detection, verification, or human approval. The fifth is *evaluation*: attack success, task utility, false refusal, cost, and transferability. The sixth is *governance*: logging, auditability, retention, disclosure, and incident response.

3.3 Direct Prompt Injection and Jailbreaks

Direct prompt attacks attempt to override policies or elicit unsafe behavior through the user channel. They include role play, obfuscation, instruction conflicts, encoding tricks, and multi-turn persuasion. Although direct jailbreaks are not the whole security problem, they reveal weaknesses in instruction following and refusal behavior. They also provide datasets and evaluation methods for testing whether models preserve higher-priority constraints under adversarial pressure [5, 15, 44, 65, 167, 168, 220, 256].

3.4 Indirect Injection Through Retrieval and Web Content

Indirect injection occurs when untrusted content enters the context through retrieval, browsing, email, documents, or tool outputs. The agent may treat this content as evidence, but an attacker can embed instructions that target the model’s control policy. Defenses require provenance labels, context separation, instruction hierarchy, sanitization, and verification of actions that originate from untrusted spans. RAG systems are particularly exposed because retrieved passages are often inserted directly into prompts [5, 15, 44, 65, 167, 168, 220, 256].

3.5 Tool and Function-Calling Security

Tool calls make prompt injection consequential. A malicious instruction may ask the agent to call an API, reveal a token, delete a record, or send data to an attacker. Function calling helps by making actions explicit and machine-checkable, but security requires more: typed permissions, argument

validation, data-flow controls, confirmation for irreversible actions, and monitors that compare tool calls against the user’s original intent. Tool-use security also depends on safe defaults and narrow scopes [5, 15, 44, 65, 167, 220, 256, 265].

3.6 Data Exfiltration and Privacy

Agents often hold secrets: conversation history, documents, API results, memory records, credentials, or business data. Prompt injection can try to move those secrets into outputs or tool arguments. Privacy research therefore intersects with agent security through prompt leakage, memorization, membership inference, data extraction, and logging policy. Strong systems track which context spans are sensitive, which tools may receive them, and whether a generated answer reveals private information [74, 143, 149, 167, 168, 194, 221, 265].

3.7 Defenses, Guardrails, and Monitoring

Defenses include prompt hardening, structured context, classifier-based detection, constitutional rules, tool sandboxes, capability scoping, information-flow checks, retrieval filtering, model-based judges, and human approval. No single layer is sufficient. Robust systems combine preventive controls before generation, runtime monitors during tool use, and audit logs after execution. Defenses must also preserve utility: overly broad refusal can make agents unusable [5, 15, 44, 65, 167, 168, 220, 256].

3.8 Evaluation and Red Teaming

Security evaluation should measure attack success over full trajectories. A benchmark may include malicious retrieved documents, conflicting instructions, sensitive context, tool permissions, and realistic user goals. Metrics should track unsafe action rate, secret leakage, false refusal, task completion, transfer across models, and robustness to adaptive attacks. Red teaming remains essential because attackers exploit interactions among components rather than isolated model responses [5, 15, 65, 167, 168, 220, 256, 265].

3.9 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit.

Prompt injection, jailbreaks, and instruction conflicts The prompt injection, jailbreaks, and instruc-

tion conflicts cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [102, 126, 128, 146, 158, 177, 179, 226, 290]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [14, 36, 103, 136, 142, 163, 219, 241, 253]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [24, 50, 64, 100, 111, 208, 237, 274, 292]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [9, 137, 172, 186, 198, 200, 247, 264, 271]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [47, 153, 155, 159, 162, 191, 218, 234, 287]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [6, 33, 42, 195, 204, 246, 281, 297, 299]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [8, 85, 119, 135].

Indirect prompt injection in retrieval and web agents

The indirect prompt injection in retrieval and web agents cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [38, 82, 167, 194, 256, 265, 276, 283, 300]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [1, 16, 25, 28, 34, 91, 199, 278, 289]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [7, 21, 32, 134, 145, 182, 223, 235, 279]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [23, 52, 94, 133, 244].

Tool-use, function-calling, and permission security

The tool-use, function-calling, and permission security cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [15, 44, 65, 83, 150, 173, 202, 277, 298]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [4, 11, 43, 46, 59–61, 108, 183]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [56, 70, 98, 165, 174, 188,

258].

Data exfiltration, privacy, and prompt leakage The data exfiltration, privacy, and prompt leakage cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [41, 71, 141, 232].

Defenses, guardrails, and monitoring The defenses, guardrails, and monitoring cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [5, 92, 99, 168, 176, 184, 201, 212, 216]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [35, 69, 101, 131, 147, 185, 229, 243, 275]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [2, 81, 89, 117, 138, 161, 171, 203, 286]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [30, 40, 76, 87, 95, 118, 170, 263, 270]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [48, 164, 175, 227].

Red teaming, benchmarks, and evaluation The red teaming, benchmarks, and evaluation cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [72, 104, 110, 143, 149, 210, 214, 220, 225]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [37, 63, 105, 109, 115, 156, 207, 255, 272]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [26, 75, 112, 154, 180, 187, 259, 288, 291]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [27, 29, 80, 84, 93, 224, 249, 252, 273]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [58, 86, 96, 113, 127, 151, 205, 231, 261]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [31, 77, 88, 97, 106, 107, 122, 144, 209]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [3, 19, 55, 78, 125, 140, 190, 197, 280]. Additional retained studies broaden the cluster with variants of interfaces, do-

main, training signals, and deployment assumptions [12, 39, 54, 67, 79, 120, 169, 239, 296]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [139].

Alignment, refusal, and policy compliance The alignment, refusal, and policy compliance cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [18, 68, 130, 181, 196, 240, 250, 293, 294].

Robustness and adversarial machine learning The robustness and adversarial machine learning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [17, 74, 123, 221, 233, 236, 260, 266, 269]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [49, 90, 114, 121, 206, 245, 248, 268, 282]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [22, 51, 66, 73, 129, 157, 228, 238, 295]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [20, 53, 62, 178, 192, 193, 217, 267, 285]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [10, 211, 213, 215, 242, 251, 254, 257, 284]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [13, 124, 132, 148, 166, 189, 222, 230, 262]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [45, 57, 116, 152, 160].

4 Challenges and Prospects

4.1 Compositional Attack Surfaces

Agent attacks compose across components: a poisoned document influences a model, which calls a tool, which writes memory, which later influences another task. Future defenses must reason over trajectories and data flow rather than checking each message independently.

4.2 Authority and Provenance

Agents need explicit authority labels for context. System policies, user requests, retrieved data, tool

observations, and model thoughts should not occupy the same instruction level. Provenance-aware prompting and execution can reduce instruction/-data confusion, but it requires support from tool schemas and memory systems.

4.3 Least Privilege for Tools

The most practical safety gains may come from narrowing tool capabilities. Agents should receive only the permissions needed for the current task, with separate read, write, execute, and transmit scopes. Irreversible actions should require confirmation or independent verification.

4.4 Balancing Safety and Utility

Security mechanisms can degrade useful behavior through false positives, excessive refusals, or high latency. The field needs evaluations that report both attack resistance and task utility, so defenses cannot win by disabling the agent.

5 Conclusion

Prompt injection and tool-use security are defining challenges for LLM agents. This survey reviewed direct and indirect prompt attacks, jailbreaks, tool-call permissions, data exfiltration, guardrails, monitoring, red teaming, and governance. The central lesson is that agent security is a systems property. Reliable agents need provenance-aware context, capability-scoped tools, sandboxed execution, runtime monitoring, and evaluations that reflect realistic trajectories over untrusted inputs.

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